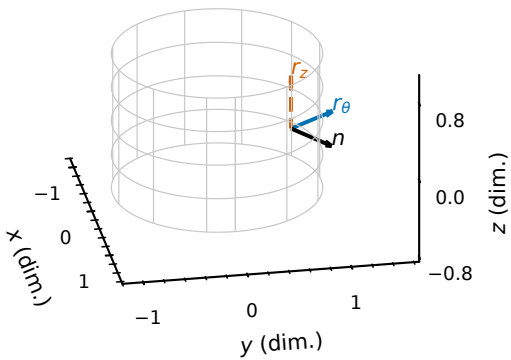
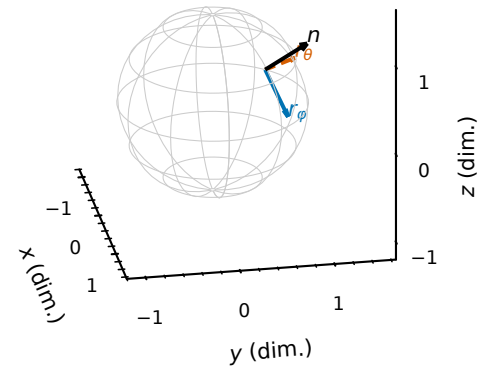
A



$r_\theta \times r_z = R\, e_r$; outward

B



$r_\varphi \times r_\theta = R^2 \sin\varphi\, e_r$; outward